

Oles Matsyshyn, Ph.D.

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RESEARCH PROFILE

Theoretical physicist working on quantum materials, nonequilibrium dynamics, nonlinear response, superconductivity, and Floquet/open quantum systems. My work combines analytical theory with large-scale numerical simulation, including light-matter interaction, quantum geometry, impurity effects, and coupling to fermionic and bosonic baths. I am particularly interested in nonequilibrium quantum phases, nonlinear optical response, superconducting response, and bridges between quantum materials and quantum-information-inspired device physics.

ACADEMIC POSITIONS AND RESEARCH EXPERIENCE

- **Nanyang Technological University** 2022–Present
Research Fellow Singapore
 - Developed theoretical and numerical frameworks for nonequilibrium quantum systems, quantum geometry, nonlinear optical response, and superconducting response.
 - Built Python/MATLAB simulation pipelines for multi-band Hamiltonians, Floquet open quantum systems, stochastic dynamics, noisy dynamics, and driven quantum materials.
 - Conducted research on graphene, moire materials, heterostructures, topological systems, PT-symmetric magnets, and superconducting quantum response.
 - Contributed to a productive research program with 12+ papers during the NTU period, including multiple Physical Review Letters publications and high-impact preprints.
- **Max Planck Institute for the Physics of Complex Systems** 2018–2022
Ph.D. Researcher; degree issued by Technische Universitat Dresden Dresden, Germany
 - Completed Ph.D. in theoretical physics with thesis awarded *magna cum laude*.
 - Worked on nonlinear response, current rectification, quantum geometry, and nonequilibrium dynamics in quantum materials.
 - Developed numerical simulations for Weyl semimetals, Dirac fermion models, and driven electronic systems.

EDUCATION

- **Max Planck Institute for the Physics of Complex Systems / TU Dresden** 2018–2022
Ph.D. in Physics Dresden, Germany
 - Thesis awarded *magna cum laude*.
- **Taras Shevchenko National University of Kyiv** 2016–2018
M.Sc. in Physics; Quantum Field Theory specialization Kyiv, Ukraine
 - Master thesis with distinction; red diploma.
- **Taras Shevchenko National University of Kyiv** 2012–2016
B.Sc. in Physics; Quantum Field Theory specialization Kyiv, Ukraine

RESEARCH AREAS

- **Quantum Dynamics and Open Systems** Python, MATLAB
Floquet systems, bath-coupled dynamics, nonequilibrium steady states
 - Developed tools for periodically driven Floquet open quantum systems coupled to fermionic and bosonic thermal baths.
 - Modeled noisy multiband quantum evolution, unitary response, Berry phases, Berry curvature, and quantum metric effects.
- **Nonlinear Response and Quantum Geometry** Theory, simulation
Nonlinear Hall effects, rectification, photocurrents, quantum metric, Berry curvature
 - Developed and simulated nonlinear optical and transport responses in quantum materials, including rectification and photovoltaic mechanisms.
 - Studied the role of quantum geometry in nonequilibrium response functions and light-driven electronic systems.
- **Superconductivity and Qubit-Relevant Architectures** Theory, simulation
Superconducting response, nonlinear electrodynamics, material-based qubit concepts
 - Developed theory and numerical simulations for nonlinear photocurrent and supercurrent response in non-centrosymmetric superconductors.
 - Explored bridges between quantum-materials physics and quantum-information-inspired superconducting/qubit architectures.
- **Machine Learning and Scientific Data Analysis** Python
Feature engineering, imputation, validation, survival analysis, quantum phase search
 - Built an end-to-end machine-learning pipeline for survival prediction in myeloid leukemia patients using real-world clinical and molecular data.
 - Applied machine-learning workflows to quantum lattice models and parameter-based phase identification.

SELECTED PUBLICATIONS AND PREPRINTS

- [J.1] O. Matsyshyn, L.-k. Shi, I. Sodemann. **High-Temperature Quantum Oscillations of a Non-equilibrium Non-Fermi Liquid.** *arXiv:2512.21763*.
- [J.2] J. Tan[†], O. Matsyshyn[†], G. Vignale, J. C. W. Song. **Nonequilibrium Exchange Nonlinear Hall Effect.** *arXiv:2512.06074*. [†]Equal contribution.
- [J.3] O. Matsyshyn, J. C. W. Song. **Superconducting Photocurrent and Light-Enriched Supercurrent Phase Relation.** *arXiv:2507.06290*.
- [J.4] F. J. Garc'ia de Abajo, *et al.*, O. Matsyshyn, *et al.* **Roadmap for Photonics with 2D Materials.** *ACS Photonics* 2025, 5c00353.
- [J.5] O. Matsyshyn, G. Vignale, J. C. W. Song. **Superconducting Berry Curvature Dipole.** *Physical Review Letters* 136, 246902.
- [J.6] L.-k. Shi[†], O. Matsyshyn[†], J. C. W. Song, I. Sodemann. **The Ultra-critical Floquet Non-Fermi Liquid.** *Physical Review Letters* 134, 196401 (2025). [†]Equal contribution.
- [J.7] O. Matsyshyn, Y. Xiong, J. C. W. Song. **Kramers dichroism in PT-symmetric magnets.** *Phys. Rev. Research* 8, 033046.
- [J.8] L.-k. Shi[†], O. Matsyshyn[†], J. C. W. Song, I. Sodemann Villadiego. **Floquet Fermi Liquid.** *Physical Review Letters* 132, 146402 (2024). Selected for the cover. [†]Equal contribution.
- [J.9] J.-X. Hu, O. Matsyshyn, J. C. W. Song. **Nonlinear Superconducting Magnetoelectric Effect.** *Physical Review Letters* 134, 026001 (2025).
- [J.10] O. Matsyshyn, J. C. W. Song, I. Sodemann Villadiego, L.-k. Shi. **The Fermi-Dirac Staircase Occupation of Floquet Bands and Current Rectification Inside the Optical Gap of Metals.** *Physical Review B* 107, 195135 (2023). Editorial Suggestion.
- [J.11] O. Matsyshyn, Y. Xiong, A. Arora, J. C. W. Song. **Layer Photovoltaic Effect in van der Waals Heterostructures.** *Physical Review B* 107, 205306 (2023).
- [J.12] L.-k. Shi, O. Matsyshyn, J. C. W. Song, I. Sodemann Villadiego. **The Berry Dipole Photovoltaic Demon and the Thermodynamics of Photo-current Generation Within the Optical Gap of Metals.** *Physical Review B* 107, 125151 (2022). Editorial Suggestion.
- [J.13] O. Matsyshyn, F. Piazza, R. Moessner, I. Sodemann. **The Rabi Regime of Current Rectification in Solids.** *Physical Review Letters* 127, 126604 (2021).
- [J.14] O. Matsyshyn, U. Dey, I. Sodemann, Y. Sun. **The Berry Phase Rectification Tensor and the Solar Rectification Vector.** *Journal of Physics D: Applied Physics* 54, 404001 (2021).
- [J.15] O. Matsyshyn, I. Sodemann. **The Non-linear Hall Acceleration and the Quantum Rectification Sum Rule.** *Physical Review Letters* 123, 246602 (2019). Cover article.
- [J.16] O. Matsyshyn, A. I. Yakimenko, E. V. Gorbar, S. I. Vilchinskii, V. V. Cheianov. **p-wave Superfluidity in Mixtures of Ultracold Fermi and Spinor Bose Gases.** *Physical Review A* 98, 043620 (2018).
- [J.17] A. I. Yakimenko, O. Matsyshyn, A. O. Oliinyk, V. M. Biloshytskyi, O. G. Chelpanova, S. I. Vilchynskii. **Analogues of Josephson Junctions and Black Hole Event Horizons in Atomic Bose–Einstein Condensates.** *Low Temperature Physics* 44, 1032 (2018). Cover article.

BOOK AND PUBLIC OUTREACH

- **Toys or Physics? Explaining Physics Through Toys** 2023
World Scientific; DOI: [10.1142/13547](https://doi.org/10.1142/13547)
 - Educational book introducing core physics concepts through everyday toys and hands-on activities.
 - Awarded the 2024 Nautilus Silver Medal in the Middle Grade Non-Fiction category.

TEACHING, MENTORING, AND ACADEMIC SERVICE

- **Reviewer for Physical Review Letters and Physical Review B** 2022–Present
- **Mentoring Ph.D. student**, Nanyang Technological University 2025
- **Examination assistance**, Nanyang Technological University 2024
- **Mentoring Master student**, Nanyang Technological University 2023
- **Condensed Matter Journal Club Lead**, Nanyang Technological University 2022–2023
- **Conference Session Chair**, Institute of Physics, Singapore 2022
- **Teaching Assistant**, Technische Universit"at Dresden 2020–2021

HONORS AND AWARDS

• Nautilus Silver Medal, Middle Grade Non-Fiction; for <i>Toys or Physics?</i>	2024
• Physical Review Letters Cover Article, <i>The Non-linear Hall Acceleration and the Quantum Rectification Sum Rule</i>	2019
• Physical Review Letters Cover Selection, <i>Floquet Fermi Liquid</i>	2024
• Physical Review B Editorial Suggestion, <i>Fermi-Dirac Staircase Occupation of Floquet Bands</i>	2023
• Physical Review B Editorial Suggestion, <i>Berry Dipole Photovoltaic Demon</i>	2022
• Ph.D. thesis awarded <i>magna cum laude</i> , TU Dresden / Max Planck Institute	2022
• Master thesis with distinction, Taras Shevchenko National University of Kyiv	2018

SKILLS

• Programming and Tools: Python, MATLAB, Git, $\LaTeX$, C++.	
• Scientific Computing: Numerical simulation, high-performance computing workflows, Monte Carlo methods, parameter sweeps, visualization, reproducible research pipelines.	
• Quantum and Condensed Matter: Quantum dynamics, open quantum systems, Floquet systems, superconductivity, quantum geometry, nonlinear response, many-body lattice models, light-matter interaction.	
• Data Science and Machine Learning: Regression, classification, clustering, PCA, XGBoost, feature engineering, imputation, survival analysis.	
• Mathematical Modeling: Perturbative and non-perturbative methods, stochastic modeling, optimization, response theory, quantum field theory.	

SELECTED CERTIFICATIONS

• Mathematical Methods for Quantitative Finance, Massachusetts Institute of Technology / edX	2025
• Fundamentals of Quantitative Modeling, Wharton	2024–2025
• Python for Business and Finance, Udemy	2024–2025

ADDITIONAL INFORMATION

Languages: English professional; Ukrainian native; Russian native; German intermediate; Romanian basic.	
Research interests: nonequilibrium quantum matter, superconductivity, Floquet systems, quantum geometry, nonlinear optical response, open quantum systems, quantum materials, quantum-information-inspired materials platforms.	

REFERENCES

Available upon request.